Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/14/2005
PI Name:	Plaut, Karen Ph.D.		
Project Title:	Is Gravity A Continuum? Investigations Of Mechanisms Mediating Gravity-Induced Change In Mammary Metabolism And Maternal Behavior In Pregnant And Lactating Rats.		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	48824-1225	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	01/01/2005	**End Date:**	07/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wade, Charles (NASA Ames) Ronca, April Ph.D. (Wake Forest University School of Medicine)		
Grant/Contract No.:	NNA05CP91A ; NCC2-1373		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Homeostasis, the dynamic state of physiological balance, is fundamentally changed in mammals exposed to either microgravity (spaceflight) or hypergravity (centrifugation). In pregnant and lactating mammals, failure to maintain homeostasis exerts pronounced effects on both mothers and their offspring. Our previous data, derived from pregnant rats exposed to either micro or hypergravity, support: 1) decreased mammary metabolic activity with increased gravitational force; 2) changes in maternal and neonatal energy balance; 3) altered maternal care; and 4) neonatal mortality. Gestation day 11 (G11) rats were placed on the 24 ft. centrifuge and exposed to 2-g hypergravity (HG), remained at 1-g as stationary controls (SC) or were pair-fed (PF) to match feed intake of the HG group. To determine metabolic activity, mammary tissue was collected from anesthetized animals, sliced 0.5 mm thick and placed in flasks containing Krebs-Ringer Buffer with 10mM glucose, 1 microg/ml insulin and 1 microCi/flask of U-14C labeled glucose. The flasks were gassed with O2, sealed with rubber stoppers containing filter paper and incubated at 37C for 3 hrs, in a shaking water bath. Metabolic activity was stopped by killing the tissue with 0.5 M H2SO4. Glucose oxidation to CO2 was determined by counting 14C-CO2 trapped on the filter papers. CO2 incorporation into lipid was measured by petroleum ether extractions and measuring in a liquid scintillation counter. Results were expressed as nmoles of U-14C glucose utilized per 100 mg tissue/3 hrs. Mammary metabolism as measured by glucose oxidation to carbon dioxide and incorporation into lipid was significantly decreased in response to hypergravity during pregnancy and lactation. 1.)To measure metabolic and hormonal changes in metabolism during hypergravity that may explain how gravitational forces alter whole body physiology during pregannncy and lactation. 2.) To measure behavioral changes that are coupled with physiological changes when animals are exposed to hypergravity and have the additional stress of pregnancy and lactation. These tasks will help us to understand the physiological regulation of metabolism in altered gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have established a very complete data set that describes changes in metabolism in mammary tissue, liver and adipose (liver and adipose funded by NASA-EPSCoR) in animals exposed to hypergravity that are pregnant or lactating. In particular this data set can be used to examine the coordinated regulation of lipid metabolism and may provide insight into obesity related disorders as well as gestational diabetes.
Task Progress:	Additional studies have been doen to determine if prolactin can over ride the lactation defect. The inufsions revealed that neither prolactin no an implan to corticosterone could override the lactation defect as measured by metabolic actvitity.
Bibliography Type:	Description: (Last Updated: 06/19/2025)
Abstracts for Journals and Proceedings	Zakrzewska E.. , Lintault, L., Maple, R., L. Baer "Zakrzewska E.. , L. Lintault, R. Maple, L. Baer, A. Ronca, C. Wade and K.Plaut. 2004. Effects of 2 x gravity (2-g) on prolactin receptor expression in adipose tissue of periparturient rats." FASEB J (submitted) , Feb-2005